THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 13: **Statistics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70, 70–80 and frequencies 3, 4, p, 4, 15, 10, 19, 22 respectively is 49. The value of p is: 1

(a) 28 **(b)** 23 **(c)** 22 **(d)** 25

2. The median of a distribution with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270 and frequencies 12, 8, x, 25, y, 18, 19 respectively, and total frequency 100, is 238.4. The value of x is: 1

(a) 10 **(b)** 8 **(c)** 9 **(d)** 11

3. For the data with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 with frequencies 23, 4, 25, 6, 3, 19, 20 respectively, The mean of the data is: 1

(a) 270 **(b)** 269.8 **(c)** 268.8 **(d)** 270.8

4. For the data with classes 20–40, 40–60, 60–80, 80–100, 100–120, 120–140, 140–160 with frequencies 3, 13, 11, 17, 15, 24, 17 respectively, The median of the data is: 1

(a) 100 **(b)** 103.6 **(c)** 108 **(d)** 120

5. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 with frequencies 17, 21, 24, 17, 6, 4, 11 respectively, The mode of the data is: 1

(a) 63 **(b)** 68 **(c)** 60 **(d)** 70

6. For a distribution with median class 20–30, $N = 50$, $cf = 18$ and $f = 14$, the median is: 1

(a) 25 **(b)** 24 **(c)** 22.5 **(d)** 28

7. The mean of a distribution with classes 100–120, 120–140, 140–160, 160–180, 180–200, 200–220, 220–240, 240–260 and frequencies 19, 3, 4, 12, p, 13, 22, 7 respectively is 184.6. The value of p is: 1

(a) 20 **(b)** 22 **(c)** 19 **(d)** 25

8. The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 20, x, 18, y, 12, 7, 18 respectively, and total frequency 100, is 25. The value of x is: 1

(a) 20 **(b)** 21 **(c)** 22 **(d)** 6

9. **Assertion (A):** For the class interval 10-20, the class mark is 15. 1

Reason (R): The class mark of a class interval is $\dfrac{\text{upper limit} + \text{lower limit}}{2}$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** For grouped data the median class is the class whose cumulative frequency is just greater than or equal to $\dfrac{n}{2}$. 1

Reason (R): The median class is the class which has the highest frequency.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The mean of a distribution with classes 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340 and frequencies 3, 24, p, 10, 15, 14, 17 respectively is 274. Find the value of p. 2

12. The marks obtained by students in a test are given below. **2**

Marks	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Number of students	3	4	23	4	15	10	19	22

Find the mode.

13. The marks obtained by students in a test are given below. **2**

Marks	200–220	220–240	240–260	260–280	280–300	300–320	320–340
Number of students	23	4	25	6	3	19	20

Find the median.

14. The mean of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110, 110–120 and frequencies 7, 23, 19, p, 9, 16, 13, 11 respectively is 78.8. Find the value of p. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The marks obtained by students in a test are given below. **3**

Marks	40–50	50–60	60–70	70–80	80–90	90–100	100–110	110–120
Number of students	11	19	8	20	23	2	13	4

Find the mean and the mode (in this order).

16. The marks obtained by students in a test are given below. **3**

Marks	40–50	50–60	60–70	70–80	80–90	90–100	100–110
Number of students	17	21	24	17	6	4	11

Find the mean.

17. The median of a distribution with classes 200–210, 210–220, 220–230, 230–240, 240–250, 250–260, 260–270 and frequencies 17, 7, x, 20, y, 19, 17 respectively, and total frequency 100, is 239.5. Find the value of x. **3**

OR

The daily wages (in ₹) of workers in a factory are given below. **3**

Daily wage (₹)	20–40	40–60	60–80	80–100	100–120	120–140	140–160
Number of workers	3	13	11	17	15	24	17

Find the mean and the mode (in this order).

18. The daily wages (in ₹) of workers in a factory are given below. **3**

Daily wage (₹)	200–220	220–240	240–260	260–280	280–300	300–320	320–340
Number of workers	10	22	24	6	21	2	15

Find the mean.

OR

The median of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 7, 10, x, 24, y, 6, 9 respectively, and total frequency 100, is 55. Find the value of x. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher recorded the time (in minutes) taken by 40 students to solve a puzzle. The classes 10-20, 20-30, 30-40, 40-50 and 50-60 had frequencies 2, 4, 10, 14 and 10 respectively. **5**

(i) Find the class mark of the class 30-40.

(ii) Write the modal class.

(iii) Find the mean time taken.

(iv) Find the mode of the data.

20. The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 20, x, 18, y, 12, 7, 18 respectively, and total frequency 100, is 25. Find the value of x. **5**

OR

The daily wages (in ₹) of workers in a factory are given below. **5**

Daily wage (₹)	20–30	30–40	40–50	50–60	60–70	70–80	80–90
Number of workers	18	13	5	16	23	10	15

Find the mean and the mode (in this order).

END OF QUESTION PAPER — Total: 40 Marks